

034IN -FONDAMENTI DI AUTOMATICA - FUNDAMENTALS OF AUTOMATIC CONTROL

FULL WRITTEN EXAMINATION - A.Y. 2023/2024

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PART 1 - Problem 1- Determining the Initial State for an LTI System Using the Input and the Output Movement - A Solution Proposal

Tag List: [STANDARD ASSESSMENT] [SYMBOLIC]

Introduction

Consider the LTI system described by the state equations

$$\begin{cases} \dot{x}_1(t) &= -x_1(t) + x_2(t) \\ \dot{x}_2(t) &= -2\,x_2(t) + u(t) \\ y(t) &= +x_1(t) - x_2(t) \end{cases}$$

Let’s indicate the system’s state at the time instant $t_0 = 0$ as the (unknown) vector x_0

$$x_0 = \begin{bmatrix} x_a \\ x_b \end{bmatrix}, \qquad x_a, x_b \in \mathbb{R}$$

Someone applied the input

$$u(t) = 15 \cdot \mathbf{1}(t)$$

to the system, starting from the instant t_0 , and recorded the complete output movement $y(t)$ (i.e., the summation of the free output and the forced output movements - refer to the class material **Part 2, pp. 30-31 and 33**). The analytical expression of the complete output movement $y(t)$ is:

$$y(t) = \left[-3 e^{-t} - e^{-2t}\right] \cdot \mathbf{1}(t)$$

Remark

As usual, $\mathbf{1}(t)$ denotes the **unit step function** (the Heaviside function): $\mathbf{1}(\tau) = \begin{cases} 0 & \longleftrightarrow \tau < 0 \\ 1 & \longleftrightarrow \tau \geq 0 \end{cases}$

Assignment

Given the input $u(t)$, the output $y(t)$ and the knowledge of the system state equations, can the initial state x_0 be determined?

Your solution has to:

- Use the Matlab's **Symbolic Toolbox**.
- Declare the symbolic variables **t** and **s**, representing respectively the time t and the complex variable s used when manipulating Laplace transforms.
- Declare the symbolic variables **xa** and **xb** and create the symbolic vector **x0**, corresponding to the initial state x_0 .
- Define the symbolic variables **u_t** and **y_t**, corresponding respectively to the input $u(t)$ and the output $y(t)$. Assign the input $u(t)$ and output $y(t)$ expressions to these variables.
- Given **y_t**, compute the output value at the instant $t = 0$ and save it in the symbolic variable **y0**.
- Compute the first time derivative of the symbolic output **y_t** with respect to the time **t** and evaluate the output's first derivative at the instant $t = 0$. Save the result in the symbolic variable **Dy0**.
- Determine the Laplace transform of the system's complete output movement. Save the resulting expression in the symbolic variable **Ys**.
- Exploiting the Laplace transform properties, in particular the so-called "**Initial Value**" **Theorem**, determine the expression of the output value at the instant $t = 0$ directly from the Laplace transform **Ys**. Assign the result to the symbolic variable **y0L**.

Note: the symbolic expression **y0L** must depend on the (unknown so far) initial state x_0 through the symbolic variables **xa** and **xb**.

- Exploiting one more time the Laplace transform properties, in particular the property regarding the **Laplace transform of the first time derivative of a causal function** $f(t)$ admitting itself the Laplace transform together with the so-called "**Initial Value**" **Theorem**, determine the expression of the first time derivative of the output $y(t)$ at the instant $t = 0$ directly from the Laplace transform **Ys**. Assign the result to the symbolic variable **Dy0L**.

Important hint: When evaluating the Laplace transform of the output time derivative $\dot{y}(t)$, remember that you have already determined the expression of the initial value of output $y(0)$ (depending on the unknowns **xa** and **xb**), so use it!

Finally:

- Compare the symbolic expressions **y0L** and **Dy0L** (both depending on the unknown values **xa** and **xb**) with the corresponding values **y0** and **Dy0**, you have already determined. Doing this comparison, you have a set of two linear equations with two unknowns (**xa** and **xb**).

- Solve the linear equations for **xa** and **xb** and finally determine the requested initial state x_0 .

Solution

```
clear variables
clc

sympref('HeavisideAtOrigin',1);

syms t real
syms xa xb real
syms s complex
syms y0L Dy0L

assumeAlso(t >=0)

% ----- ICs -----
x0 = [xa;xb];

% ---- input ---
u_t = sym(15);
Us = laplace(u_t, t, s);
% ---- output ---
y_t = -3*exp(-t)-exp(-2*t);

y0 = subs(y_t, t, 0)
```

$y_0 = -4$

```
% ----- the time derivatives of the function y(t) -----
dot_y = diff(y_t,t); % the 1st derivative of y

Dy0 = subs(dot_y, t, 0)
```

$Dy_0 = 5$

```
% -----
A = sym([-1. 1;0, -2]);
B = sym([0;1]);
C = sym([1, -1]);
D = sym(0);

sIA = s*eye(2)-A;
sIA1=inv(sIA);
Gs=collect(simplify(C*sIA1*B+D),s)
```

$G_s =$

$$\frac{-s}{s^2 + 3s + 2}$$

```
YFs = Gs*Us;      % the forced output movement
YLs = C*sIA1*x0; % the free output movement
```

```
Ys = collect(YLs+YFs,s)
```

```
Ys =
```

$$\frac{(x_a - x_b)s + 2x_a - 15}{s^2 + 3s + 2}$$

```
% -- Initial Value Theorem
collect(s*Ys,s)
```

```
ans =
```

$$\frac{(x_a - x_b)s^2 + (2x_a - 15)s}{s^2 + 3s + 2}$$

```
y0L = limit(s*Ys, s, inf)
```

$$y0L = x_a - x_b$$

```
% Laplace transform of the first derivative
Dys = collect(s*Ys-y0L, s)
```

```
Dys =
```

$$\frac{(3x_b - x_a - 15)s + 2x_b - 2x_a}{s^2 + 3s + 2}$$

```
% at t=0
Dy0L = limit(s*Dys, s, inf)
```

$$Dy0L = 3x_b - x_a - 15$$

```
EQ_SET = [y0L == y0; Dy0L == Dy0]
```

```
EQ_SET =
```

$$\begin{pmatrix} x_a - x_b = -4 \\ 3x_b - x_a - 15 = 5 \end{pmatrix}$$

```
x0SOL = solve(EQ_SET)
```

```
x0SOL = struct with fields:
    xa: 4
    xb: 8
```

The initial state x_0 is

```
x0IC = [x0SOL.xa;
        x0SOL.xb]
```

$x_{0IC} = \binom{4}{8}$